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Course/Code: DSA0604 - DATA HANDLING AND VISUALIZATION

Date: 22/07/26

SET 6: PRODUCT SALES ANALYSIS

1) Grouped Bar Chart – Q1 Sales per Product

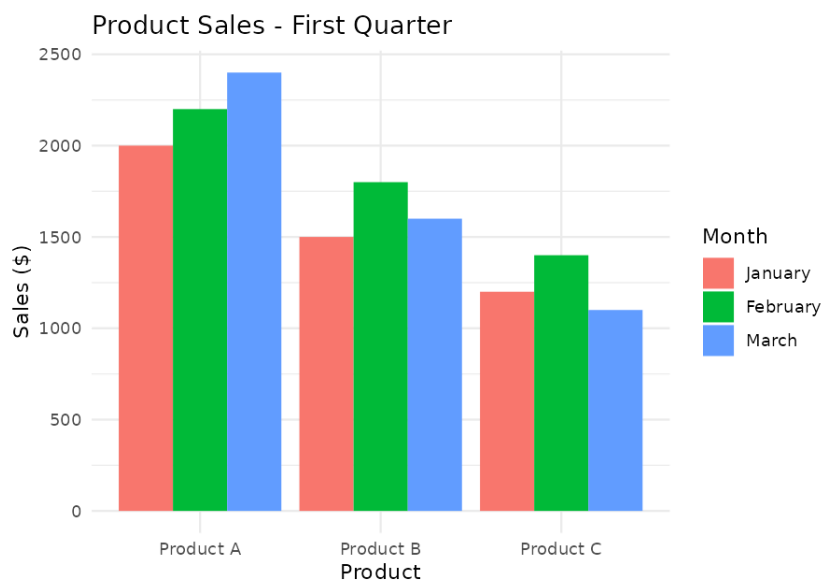
CODE:

```
library(ggplot2)
library(reshape2)

sales6 <- data.frame(
  ProductID = 1:3,
  ProductName = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
long6 <- melt(sales6, id.vars = c("ProductID","ProductName"),
  variable.name = "Month", value.name = "Sales")

ggplot(long6, aes(x = ProductName, y = Sales, fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Product Sales - First Quarter",
    x = "Product", y = "Sales ($)") +
  theme_minimal()
```

OUTPUT:



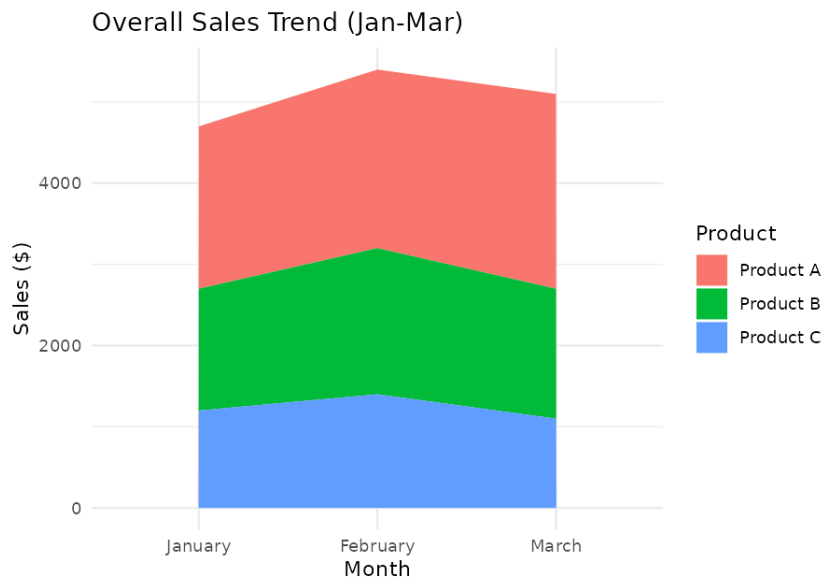
2) Stacked Area Chart – Overall Sales Trend

CODE:

```
library(ggplot2)
```

```
ggplot(long6, aes(x = Month, y = Sales, fill = ProductName, group = ProductName)) +  
  geom_area(position = "stack") +  
  labs(title = "Overall Sales Trend (Jan-Mar)",  
       x = "Month", y = "Sales ($)", fill = "Product") +  
  theme_minimal()
```

OUTPUT:



3) Table – Monthly Sales Data per Product

CODE:

```
print(sales)
```

OUTPUT:

ProductID	ProductName	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

4) Dashboard – Grouped Bar + Stacked Area + Table

CODE:

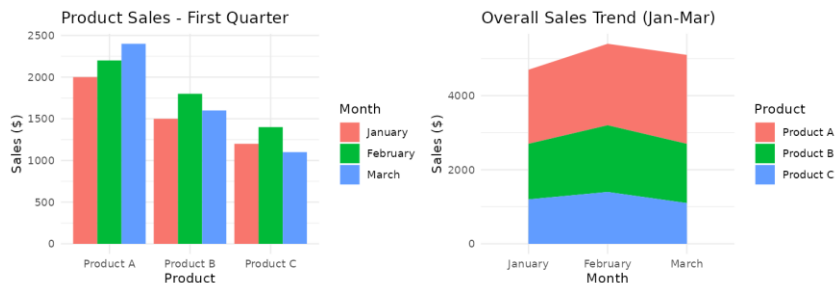
```
sales_long <- melt(sales,
  id.vars = "Product",
  variable.name = "Month",
  value.name = "Sales")

ggplot(sales_long,
  aes(Product, Sales, fill = Month)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(title = "Monthly Product Sales")

ggplot(sales_long,
  aes(Month, Sales,
    fill = Product,
    group = Product)) +
  geom_area() +
  labs(title = "Overall Sales Trend")
```

```
print(sales)
```

OUTPUT:



ProductID	ProductName	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

SET 7: CUSTOMER DEMOGRAPHICS ANALYSIS

1) Bar Chart – Distribution of Customer Ages

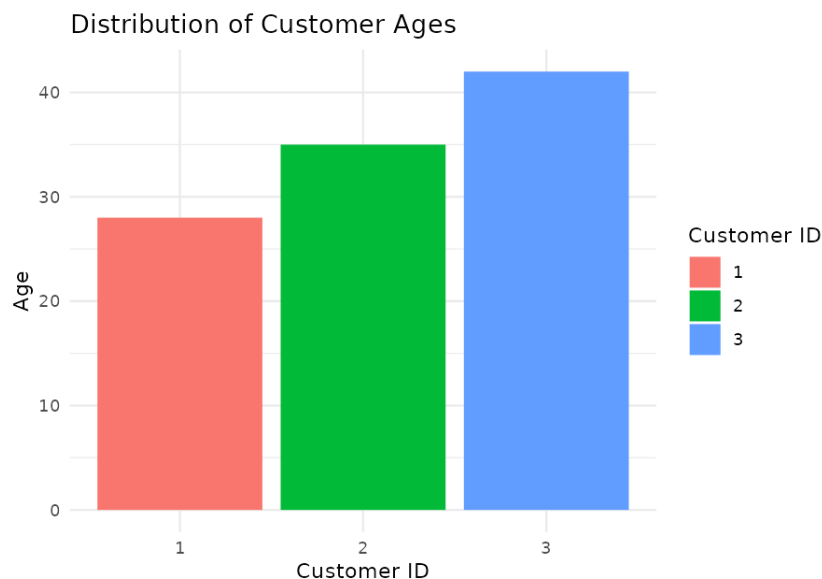
CODE:

```
library(ggplot2)
```

```
demo <- data.frame(  
  CustomerID = 1:3,  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)
```

```
ggplot(demo, aes(x = factor(CustomerID), y = Age, fill = factor(CustomerID))) +  
  geom_bar(stat = "identity") +  
  labs(title = "Distribution of Customer Ages",  
       x = "Customer ID", y = "Age", fill = "Customer ID") +  
  theme_minimal()
```

OUTPUT:



2) Pie Chart – Distribution by Gender

CODE:

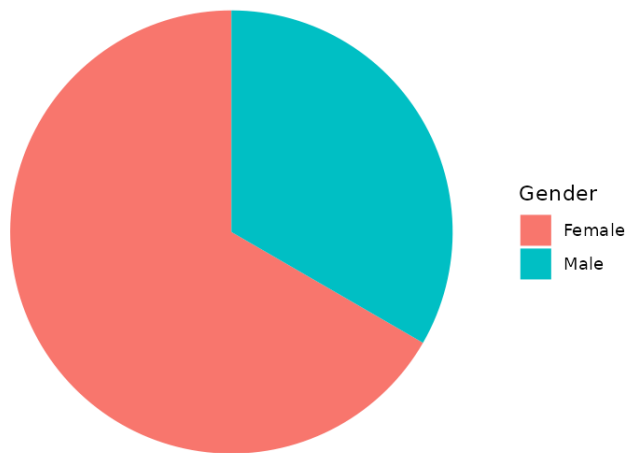
```
library(ggplot2)

gender_counts <- as.data.frame(table(demo$Gender))
colnames(gender_counts) <- c("Gender","Count")

ggplot(gender_counts, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Customer Distribution by Gender") +
  theme_void()
```

OUTPUT:

Customer Distribution by Gender



3) Table – Demographic Information per Customer

CODE:

```
print(customer)
```

OUTPUT:

CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

4) Dashboard – Bar + Pie + Table

CODE:

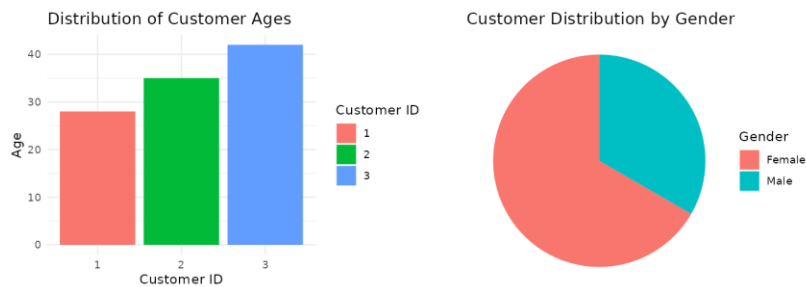
```
ggplot(customer, aes(x = factor(ID), y = Age, fill = factor(ID))) +  
  geom_bar(stat = "identity") +  
  labs(title = "Customer Age Distribution")
```

```
gender <- table(customer$Gender)
```

```
pie(gender,  
  labels = names(gender),  
  main = "Gender Distribution",  
  col = c("pink", "lightblue"))
```

```
print(customer)
```

OUTPUT:



CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

SET 8: EMPLOYEE PERFORMANCE ANALYSIS

1) Line Chart – Employee Performance Trend

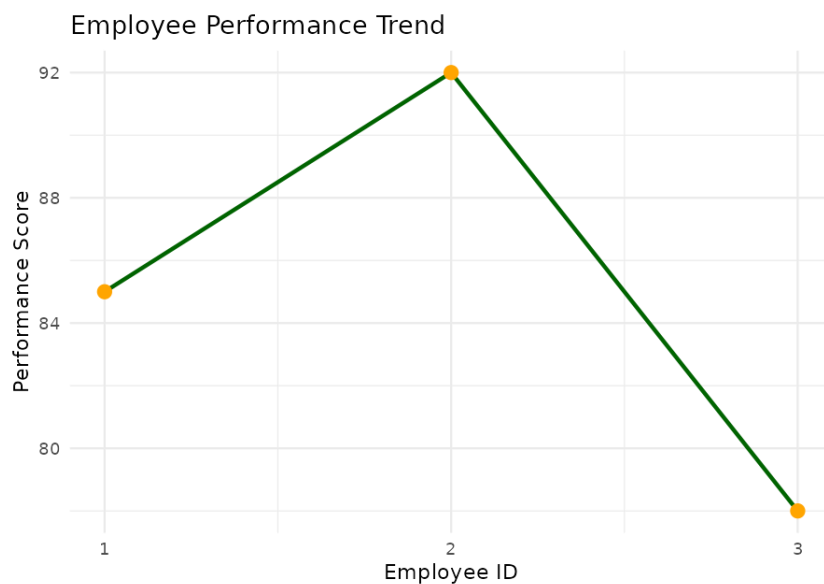
CODE:

```
library(ggplot2)
```

```
emp8 <- data.frame(  
  EmployeeID = 1:3,  
  Department = c("Sales", "HR", "Marketing"),  
  YearsOfService = c(5, 3, 7),  
  PerformanceScore = c(85, 92, 78)  
)
```

```
ggplot(emp8, aes(x = EmployeeID, y = PerformanceScore, group = 1)) +  
  geom_line(color = "darkgreen", linewidth = 1) +  
  geom_point(color = "orange", size = 3) +  
  scale_x_continuous(breaks = emp8$EmployeeID) +  
  labs(title = "Employee Performance Trend",  
       x = "Employee ID", y = "Performance Score") +  
  theme_minimal()
```

OUTPUT:



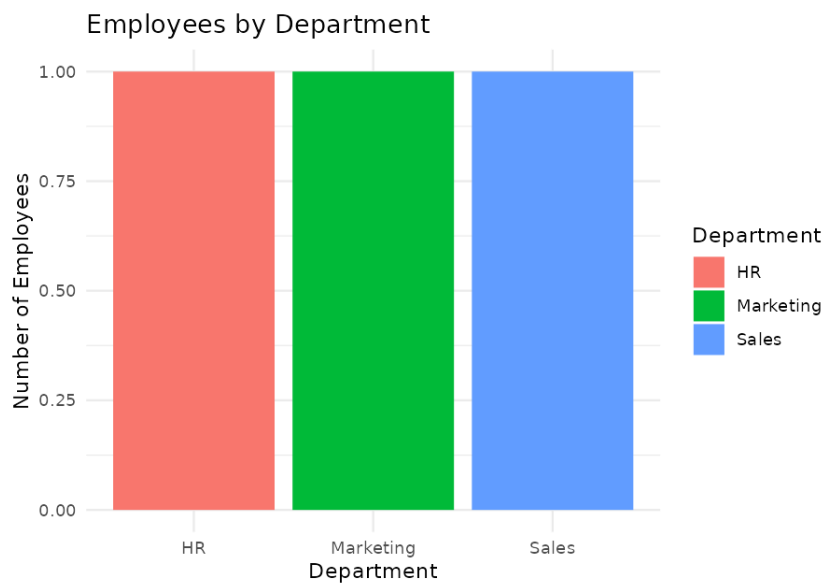
2) Bar Chart – Employees by Department

CODE:

```
library(ggplot2)

ggplot(emp8, aes(x = Department, fill = Department)) +
  geom_bar() +
  labs(title = "Employees by Department",
       x = "Department", y = "Number of Employees") +
  theme_minimal()
```

OUTPUT:



3) Table – Performance Data per Employee

CODE:

```
print(emp)
```

OUTPUT:

EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

4) Dashboard – Line + Bar + Table

CODE:

```
ggplot(emp, aes(x = YearsService, y = Performance)) +  
  geom_line(color = "blue", linewidth = 1) +  
  geom_point(size = 3, color = "red") +  
  labs(title = "Employee Performance Trend")
```

```
ggplot(emp, aes(x = Department, fill = Department)) +  
  geom_bar() +  
  labs(title = "Employees by Department")
```

```
print(emp)
```

OUTPUT:



EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

SET 9: ONLINE LEARNING ACTIVITY ANALYSIS

1) Histogram & Boxplot – Quiz_Score Distribution

CODE:

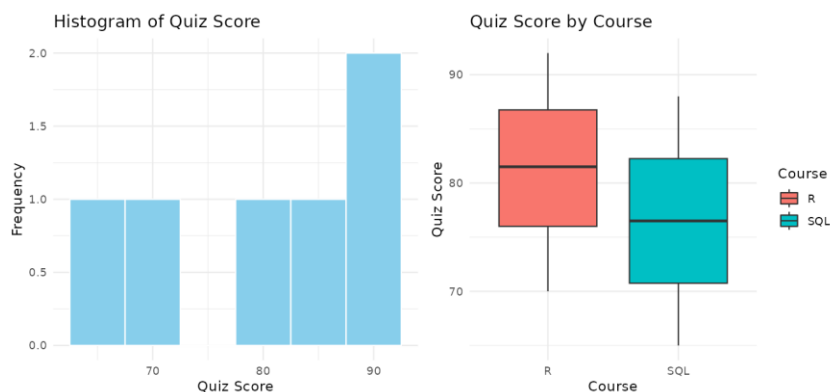
```
library(ggplot2)

learn <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = as.Date(c("2025-01-05","2025-01-05","2025-02-08",
    "2025-02-08","2025-03-12","2025-03-12"))
)

ggplot(learn, aes(Quiz_Score)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "white") +
  labs(title = "Histogram of Quiz Score", x = "Quiz Score", y = "Frequency") +
  theme_minimal()

ggplot(learn, aes(x = Course, y = Quiz_Score, fill = Course)) +
  geom_boxplot() +
  labs(title = "Quiz Score by Course", x = "Course", y = "Quiz Score") +
  theme_minimal()
```

OUTPUT:



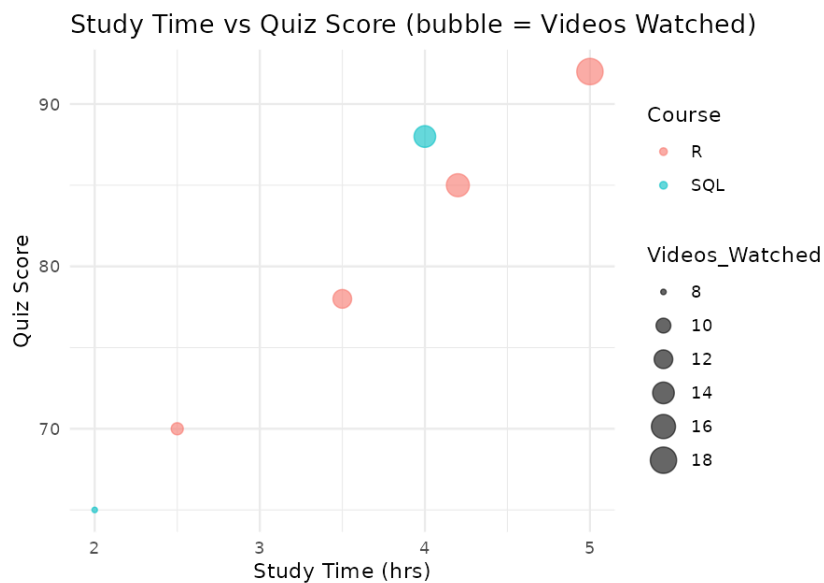
2) Scatter Plot – Study_Time vs Quiz_Score (Bubble = Videos_Watched)

CODE:

```
library(ggplot2)

ggplot(learn, aes(x = Study_Time, y = Quiz_Score,
                  size = Videos_Watched, color = Course)) +
  geom_point(alpha = 0.6) +
  labs(title = "Study Time vs Quiz Score (bubble = Videos Watched)",
       x = "Study Time (hrs)", y = "Quiz Score") +
  theme_minimal()
```

OUTPUT:



3) Line Chart – Average Quiz Score per Month with Moving Average

CODE:

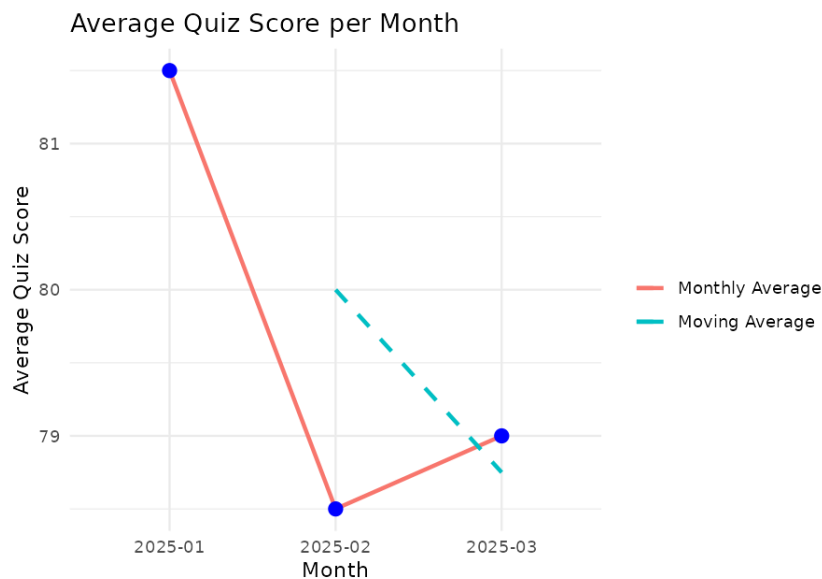
```
library(dplyr)
library(zoo)
library(ggplot2)

learn$Month <- format(learn$Login_Date, "%Y-%m")
monthly <- learn %>%
  group_by(Month) %>%
  summarise(AvgQuizScore = mean(Quiz_Score)) %>%
  arrange(Month)

monthly$MovingAvg <- rollmean(monthly$AvgQuizScore, k = 2,
                              fill = NA, align = "right")

ggplot(monthly, aes(x = Month, group = 1)) +
  geom_line(aes(y = AvgQuizScore, color = "Monthly Average"), linewidth = 1) +
  geom_point(aes(y = AvgQuizScore), color = "blue", size = 3) +
  geom_line(aes(y = MovingAvg, color = "Moving Average"),
            linewidth = 1, linetype = "dashed", na.rm = TRUE) +
  labs(title = "Average Quiz Score per Month",
       x = "Month", y = "Average Quiz Score", color = "") +
  theme_minimal()
```

OUTPUT:



SET 10: SURVEY RESPONSES ANALYSIS

1) Grouped Bar Chart – Distribution of Answers for Question 1

CODE:

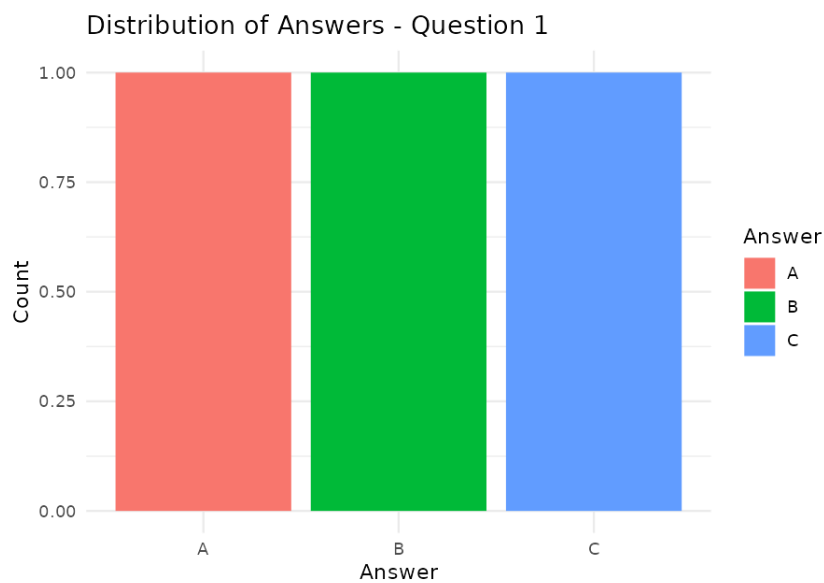
```
library(ggplot2)

survey <- data.frame(
  SurveyID = 1:3,
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

q1_counts <- as.data.frame(table(survey$Question1))
colnames(q1_counts) <- c("Answer","Count")

ggplot(q1_counts, aes(x = Answer, y = Count, fill = Answer)) +
  geom_bar(stat = "identity") +
  labs(title = "Distribution of Answers - Question 1",
       x = "Answer", y = "Count") +
  theme_minimal()
```

OUTPUT:



2) Stacked Bar Chart – Overall Distribution of Responses

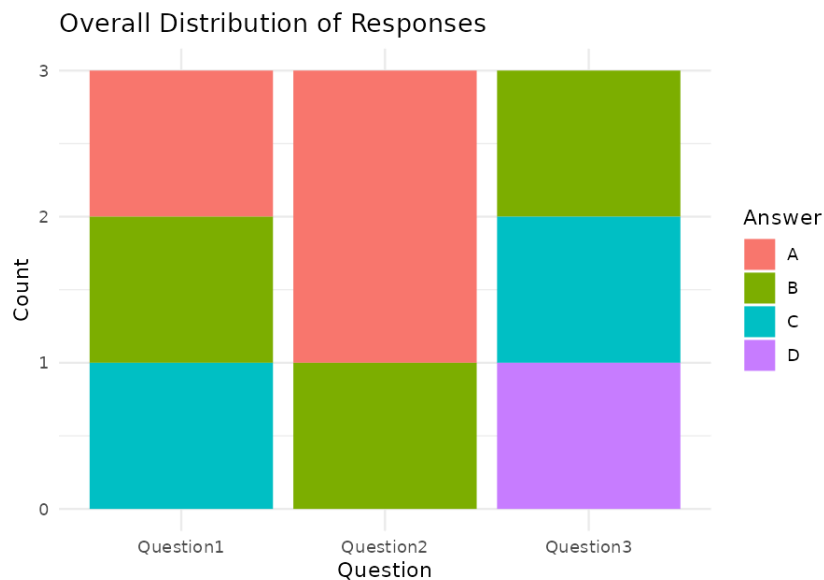
CODE:

```
library(ggplot2)
library(reshape2)

long_survey <- melt(survey, id.vars = "SurveyID",
                    variable.name = "Question", value.name = "Answer")

ggplot(long_survey, aes(x = Question, fill = Answer)) +
  geom_bar(position = "stack") +
  labs(title = "Overall Distribution of Responses",
       x = "Question", y = "Count") +
  theme_minimal()
```

OUTPUT:



3) Table – Survey Response Data per Survey

CODE:

```
print(survey)
```

OUTPUT:

SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

4) Dashboard – Grouped Bar + Stacked Bar + Table

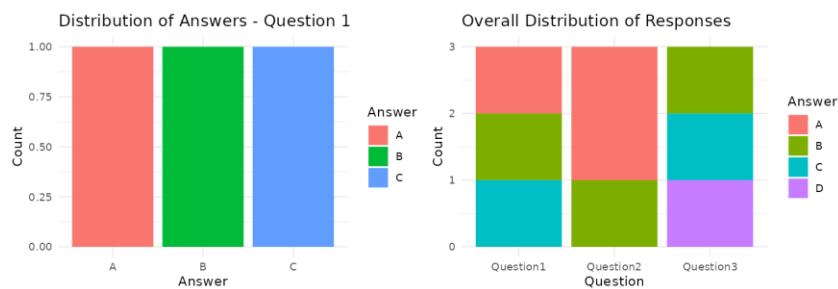
CODE:

```
ggplot(subset(survey_long, Question == "Q1"),  
       aes(x = Answer, fill = Answer)) +  
  geom_bar() +  
  labs(title = "Question 1 Responses")
```

```
ggplot(survey_long,  
       aes(x = Question, fill = Answer)) +  
  geom_bar() +  
  labs(title = "Overall Survey Responses")
```

```
print(survey)
```

OUTPUT:



SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B